

Milestone 4 Report: Incident Type Analysis

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1. Introduction

Disaster analysis plays a vital role in understanding patterns, identifying high-risk areas, and improving disaster management strategies. After performing data cleaning, temporal analysis, and geographic analysis in previous milestones, this stage focuses on analyzing disaster incident types in detail.

The primary goal of this milestone is to study the distribution of disaster types, understand how they vary across different states, and analyze their relationship with disaster assistance programs. This analysis helps in identifying dominant disaster categories and understanding their impact on society and infrastructure.

2. Objectives

The objectives of this milestone are:

- To analyze the distribution of different disaster incident types
- To identify the most frequent disaster categories
- To examine how disaster types vary across different states
- To analyze the relationship between disaster types and assistance programs

- To derive meaningful insights by integrating previous milestone findings

3. Dataset Description

The dataset used in this milestone is the cleaned and processed FEMA disaster dataset (usnd_cleaned.csv), prepared in earlier stages of the project.

Key Columns Used:

- **incidentType**: Represents the type of disaster (e.g., Flood, Storm, Hurricane)
- **state**: Indicates the state where the disaster occurred
- **year**: Year of disaster occurrence
- **month**: Month of disaster occurrence
- **ihProgramDeclared**: Indicates whether Individual Housing assistance was provided
- **paProgramDeclared**: Indicates whether Public Assistance was provided

The dataset is structured and validated, ensuring accurate and meaningful analysis.

4. Methodology

The analysis in this milestone was carried out in three major stages:

4.1 Incident Type Distribution Analysis

The dataset was grouped by **incidentType**, and the number of occurrences for each disaster type was calculated. This helped in identifying the most frequent disasters.

A **bar chart** was used to visualize the distribution, and a **pie chart** was optionally created to highlight the top disaster types.

4.2 State vs Incident Type Analysis

To understand how disasters vary across regions, the dataset was grouped by **state and incidentType**. The counts for each combination were calculated.

Two visualizations were used:

- **Stacked Bar Chart** to show how different disaster types contribute within each state
- **Heatmap** to represent the intensity of disaster occurrences across states and types

4.3 Disaster Assistance Program Analysis

The relationship between disaster types and assistance programs was analyzed using the following columns:

- **ihProgramDeclared** (Individual Housing Assistance)
- **paProgramDeclared** (Public Assistance)

The data was grouped by **incidentType**, and the total number of assistance declarations was calculated. A **grouped bar**

chart was used to compare assistance programs across disaster types.

5. Incident Type Distribution Analysis

Findings:

- Floods and storms are the most frequently occurring disaster types.
- A small number of disaster categories dominate the dataset.
- Rare disaster types such as earthquakes and volcanic events occur infrequently.

Interpretation:

The dominance of floods and storms indicates that these disasters have a widespread impact and occur more frequently due to climatic and environmental factors. The uneven distribution suggests that disaster occurrences are concentrated in specific categories rather than being evenly spread.

6. State vs Incident Type Analysis

Findings:

- Certain states experience a higher frequency of specific disaster types.
- Coastal regions tend to experience more floods and hurricanes.

- Inland regions are more prone to storms and severe weather conditions.
- Heatmap visualization clearly highlights regional disaster patterns.

Interpretation:

The variation in disaster types across states indicates that geographic and climatic conditions play a major role in determining the type of disasters. Each state has its own dominant disaster type, influenced by environmental factors.

7. Disaster Assistance Program Analysis

Findings:

- Major disasters such as floods and hurricanes trigger assistance programs more frequently.
- Public Assistance (PA) programs are activated more often than Individual Housing (IH) programs.
- Severe disasters require both types of assistance.
- Minor disasters result in fewer assistance declarations.

Interpretation:

The activation of assistance programs depends on the severity and impact of disasters. Public assistance focuses on infrastructure recovery, while individual assistance supports affected households. The higher usage of public assistance indicates the large-scale impact of disasters.

8. Integration with Previous Milestones

The findings from this milestone were combined with insights from earlier stages:

- From **Temporal Analysis**, it was observed that disaster frequency is increasing over time.
- From **Geographic Analysis**, certain states were identified as disaster-prone regions.
- From **Incident Type Analysis**, floods and storms were found to be the most common disasters.

Combined Insight:

Disasters are influenced by time, location, and type. Certain disaster types dominate specific regions and occur more frequently during particular periods.

9. Key Insights

- Floods and storms are the most dominant disaster types.
- Disaster types vary significantly across different states.
- Geographic location plays a crucial role in determining disaster type.
- Major disasters trigger higher levels of assistance programs.
- Public assistance is more commonly used than individual housing assistance.

10. Conclusion

This milestone provides a detailed understanding of disaster incident types and their distribution across states and assistance programs.

The analysis reveals that disaster occurrences are not random but follow clear patterns influenced by geographic and environmental factors. Certain disaster types dominate the dataset, and their impact requires significant government support.

Overall, this analysis highlights the importance of understanding disaster types for effective planning, resource allocation, and disaster management strategies.